

Lab 5

Static Data Members / Functions in OOP (Java - NetBeans)

Objective

- To understand the concept of static variables and static methods in Java.
- To learn how static members are shared among all objects of a class.
- To implement programs using static data members and functions.

Theory

- A static data member is shared among all objects of a class and is declared using the keyword static.
- Only one copy of a static variable exists regardless of the number of objects created.
- A static method belongs to the class rather than an object and can be called using the class name.
- Static methods can directly access only static variables.
- Static members are useful for representing common properties shared by all objects.

Syntax

```
class ClassName {  
  
    // Static Variable  
    static dataType variableName;  
  
    // Static Method  
    static void methodName() {  
        // code  
    }  
}
```

```
// Accessing static members  
ClassName.variableName;  
ClassName.methodName();
```

Example: Static Members (Java)

```
class Student {
    String name;
    int rollNumber;
    static String university = "ABC University";

    void displayInfo() {
        System.out.println("Name: " + name);
        System.out.println("Roll Number: " + rollNumber);
        System.out.println("University: " + university);
    }

    static void changeUniversity(String newUni) {
        university = newUni;
    }
}

public class StudentMain {
    public static void main(String[] args) {

        Student s1 = new Student();
        s1.name = "Ali";
        s1.rollNumber = 101;

        Student s2 = new Student();
        s2.name = "Sara";
        s2.rollNumber = 102;

        s1.displayInfo();
        s2.displayInfo();

        Student.changeUniversity("XYZ University");

        System.out.println("\nAfter Changing University:");

        s1.displayInfo();
        s2.displayInfo();
    }
}
```

Student Tasks

1. Create a Student class with a static variable university and display student details.
2. Create a Counter class with a static variable to count number of objects created.
3. Create an Employee class with static company name shared among all employees.
4. Create a BankAccount class with static interest rate and display account details.
5. Create a Book class with static library name shared by all books.